Introduction
Multicast Addressing
IGMP
MBONE
Multicast Routing

Multicast

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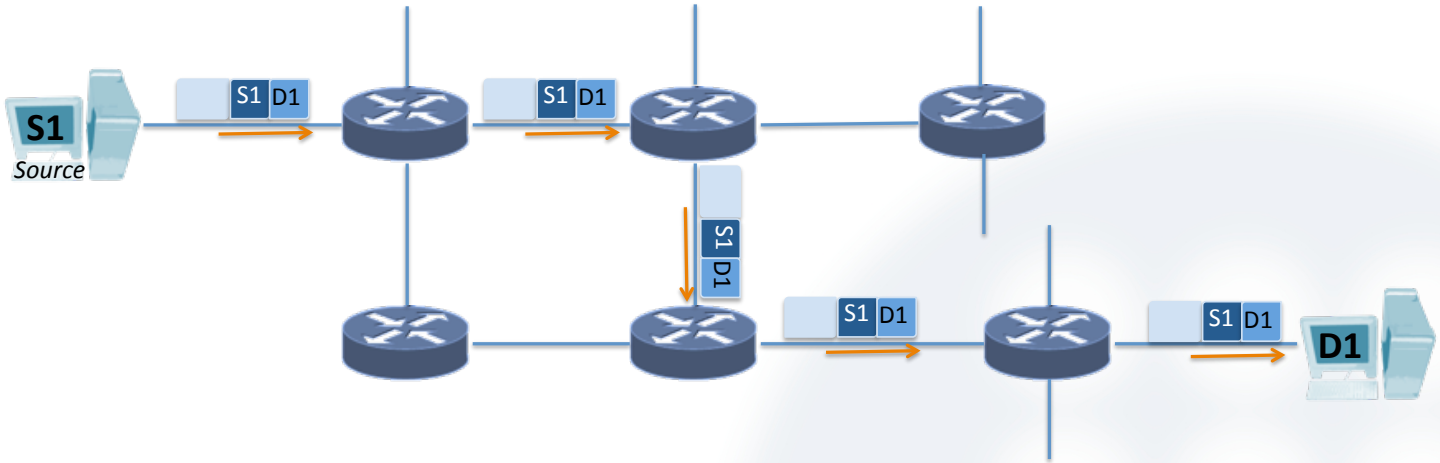
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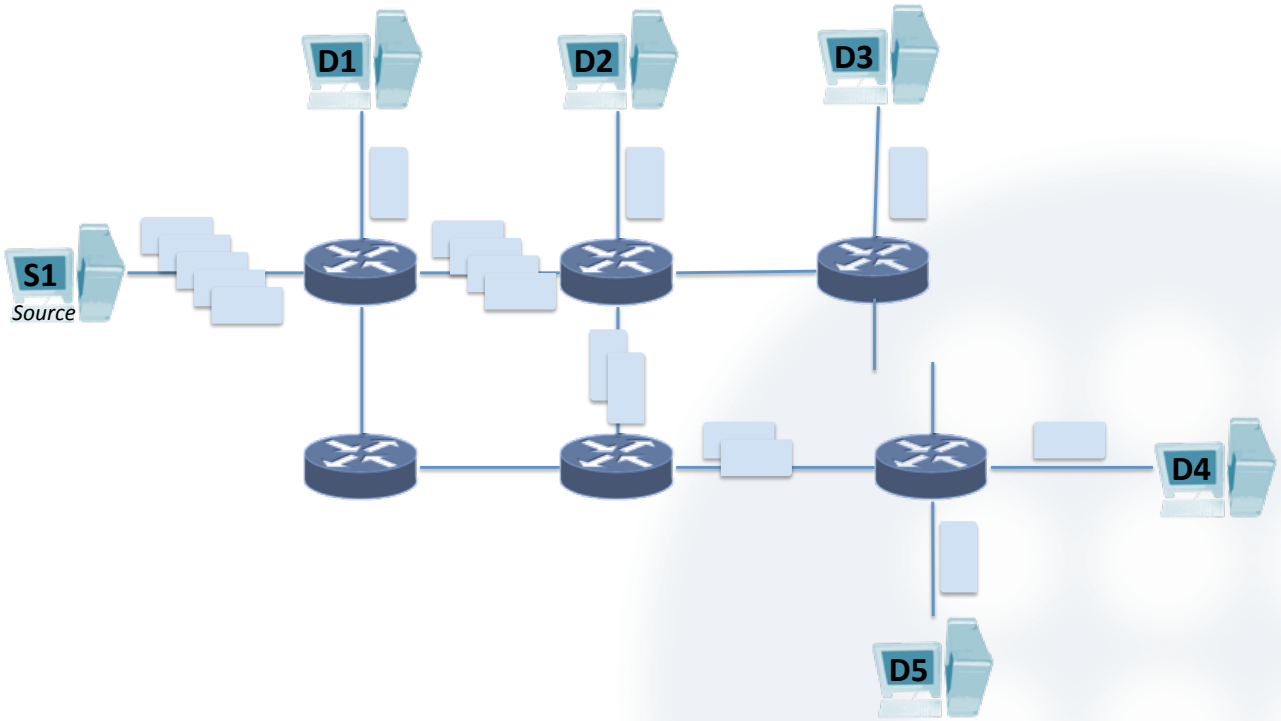
What is Unicasting?



- In unicasting, there is a single sender (source) and a single receiver (destination).
- In unicast routing, the router forwards the received packet through only one of its interfaces.

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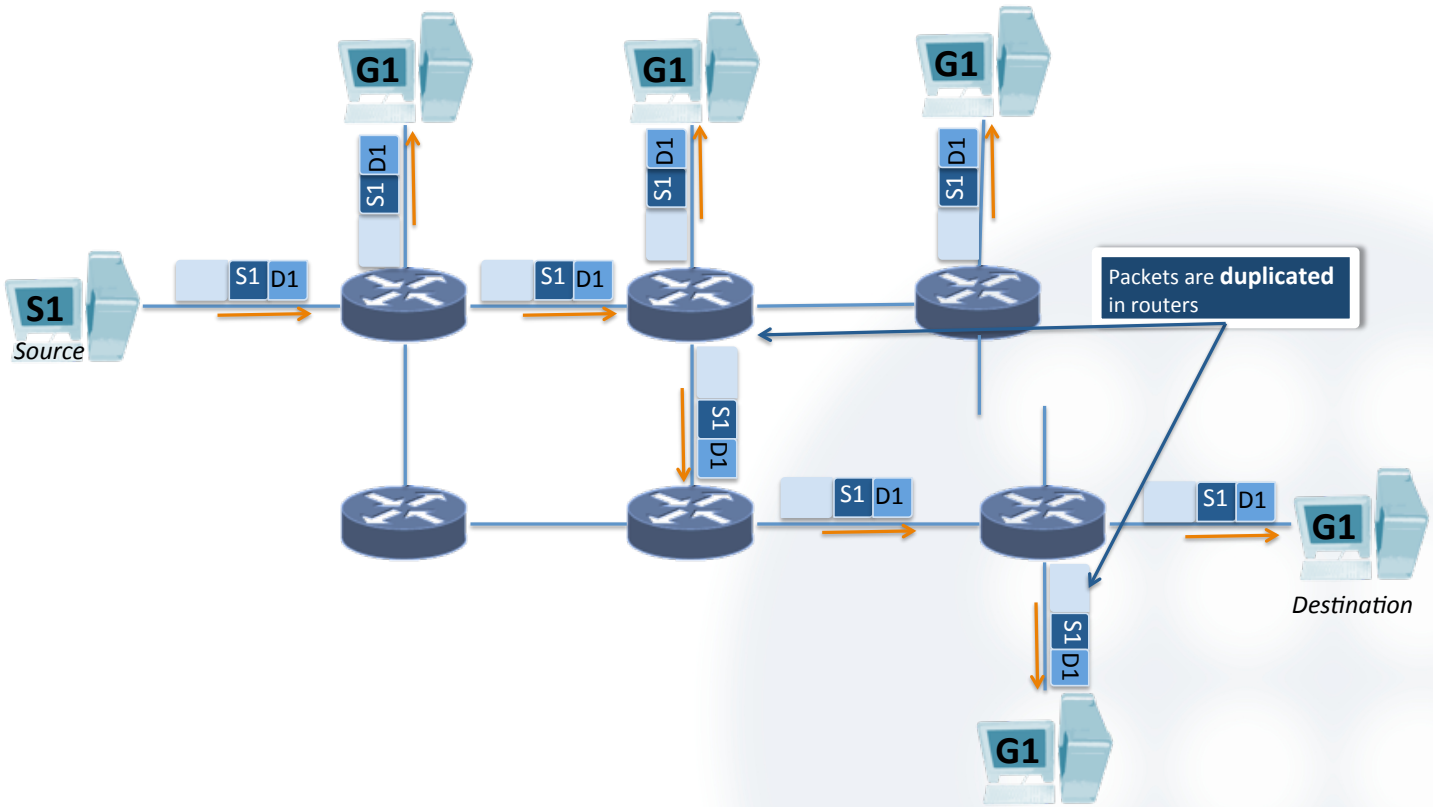
What is Multiple Unicasting?



- There is a single sender and multiple receivers.
- Emulate multicasting through multiple unicast transmission.

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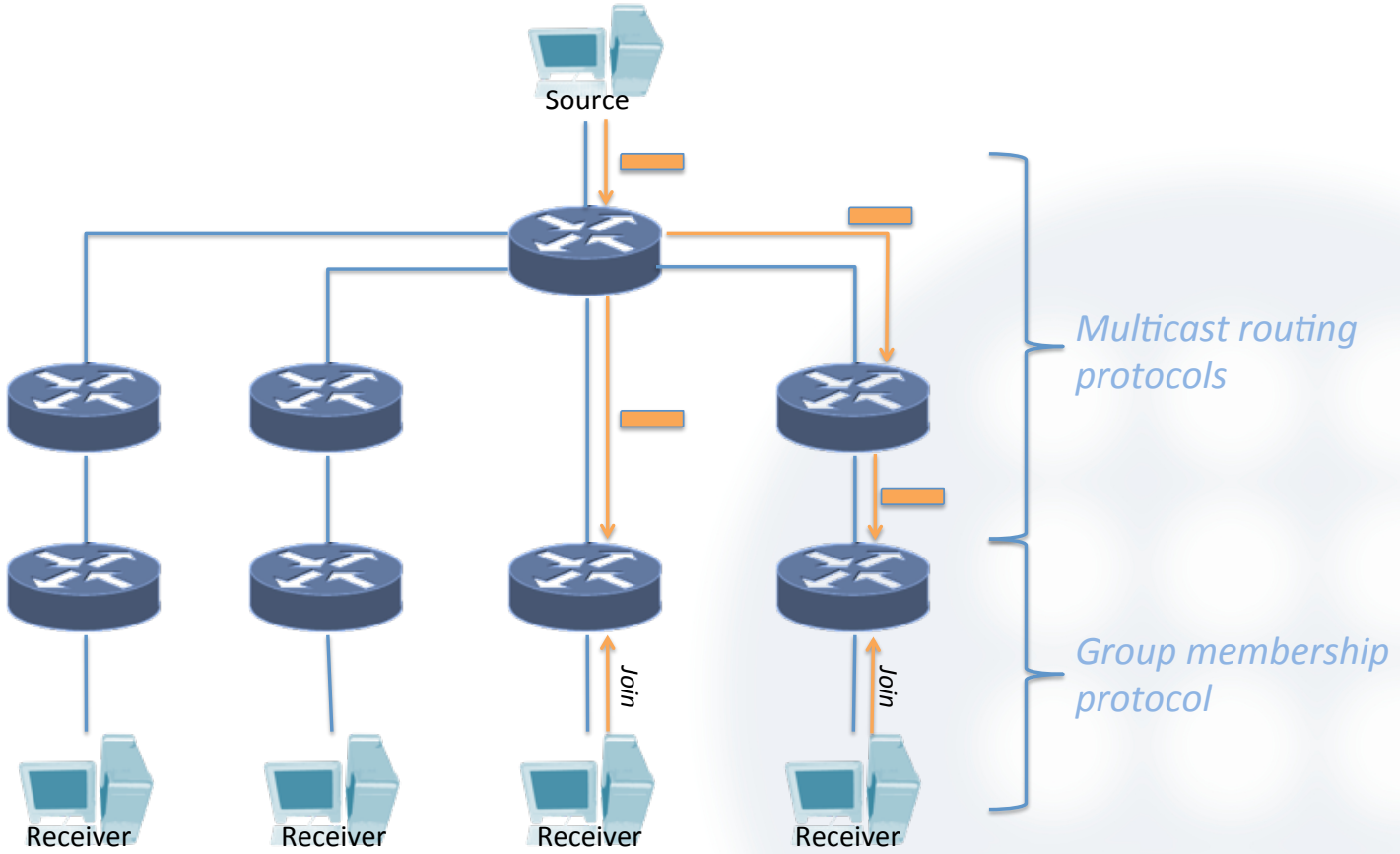
What is Multicasting?



- In multicasting, there is at least one sender and several receivers (group of receivers called multicast group).
- In multicast routing, the router may forward the received packet through several of its interfaces.

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Multicast Fordwarding



- Multicast is not broadcast.
- Receivers need to request forwarding.

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IP Multicast Service

Sending: using normal IP-Send operation, with an IP multicast address specified as the destination

- Specify outgoing network interface
- Specify IP time-to-live (TTL) on outgoing packet
- Enable/disable loop-back if the sending host is/isn't a member of the destination group on the outgoing interface

Receiving: receiving multicast packets for joined groups via normal IP-Receive operation

- Join-IP-Multicast-Group (group-address, interface)
- Leave-IP-Multicast-Group (group-address, interface)

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Applications of Multicasting I

It is estimated that thousands of groups are present at any time in the Internet. This number is increasing rapidly.
One-to-many:

- Scheduled audio-video distribution (lectures, business TV, meetings,...)
- Push media (news headlines, weather updates, radio broadcast)
- File distribution and caching (web site content, file-based updates)
- Announcements (network time, session schedules)
- Monitoring (stock prices, telemetry)

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Applications of Multicasting II

Many-to-many:

- Multimedia conferencing (audio/video, whiteboards)
- Synchronized resources (shared distributed databases)
- Concurrent processing (distributed parallel processing)
- Collaboration (shared document editing)
- Distance learning (one-to-many + feedback)
- Chat groups
- Distributed interactive simulations
- Multiplayer games
- Jam sessions (shared encoded music)

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Multicast Apps Characteristics

- Number of (simultaneous) senders to the group
- The size of the groups: number of members (receivers)
- Geographic extent or scope
- Diameter of the group measured in router hops
- Number of aggregate packets/second
- The peak/average used by source
- Level of human interactivity: lecture mode vs interactive
- Data-only (e.g. database replication) vs multimedia

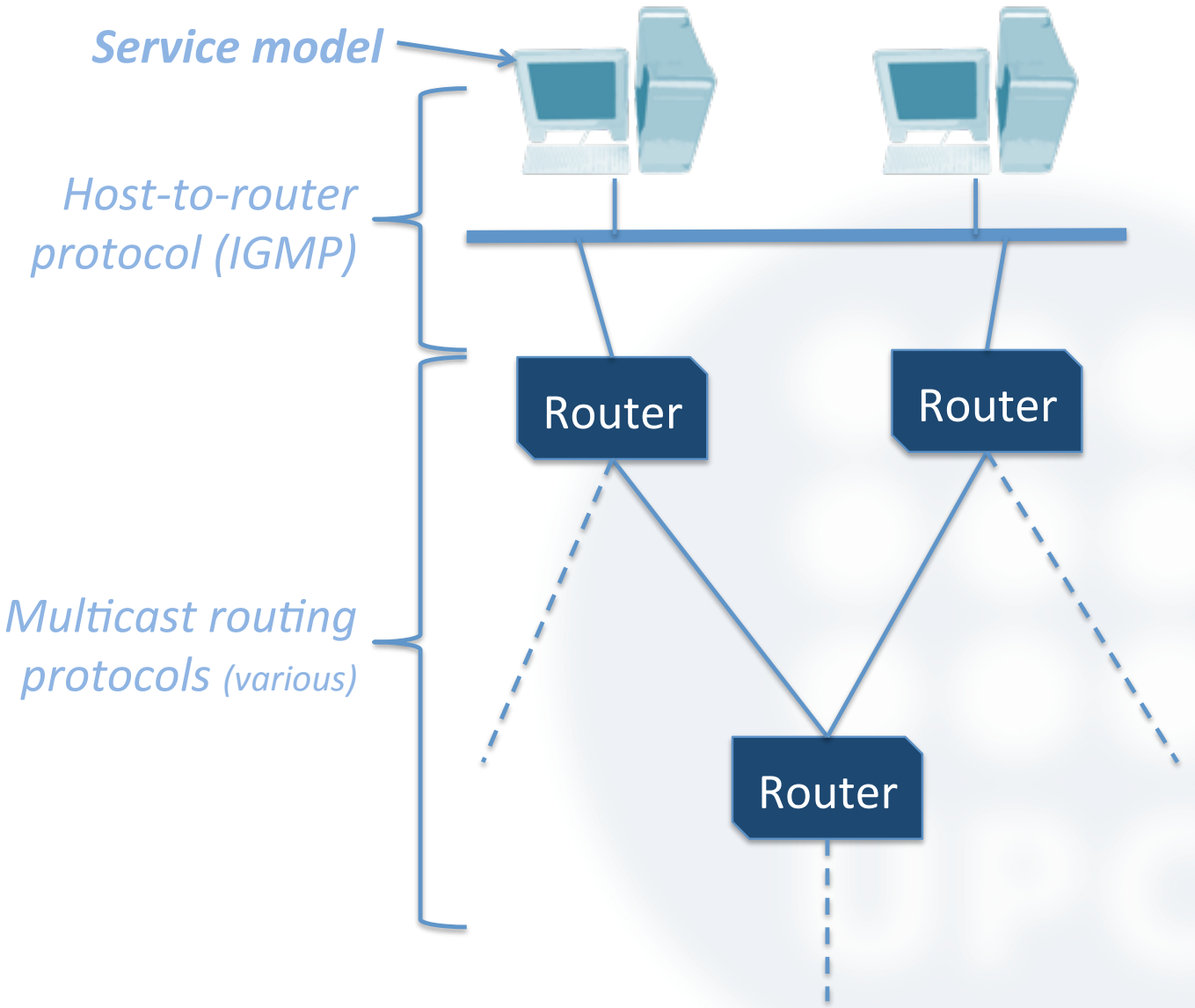
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Reliability during transport

- Unreliable multicast: the arrival of the correct message at each process is not guaranteed.
- Reliable multicast: guarantees that each message is eventually delivered in a non-corrupted form to each process in the group.
  - It does not put any restriction of the order the messages delivered.

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Link-Layer Transmission/Reception

Transmission:

- An IP multicast packets is transmitted as a link-layer multicast, on those links that support multicast
- The link-layer destination address is determined by an algorithm specific to the type of link

Reception:

- Necessary steps are taken to receive desired multicasts on a particular link, such as modifying address reception filters on LAN interfaces
- Multicast routers must be able to receive all IP multicasts on a link, without knowing in advance which groups will be used

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IP Address Structure

- The IP address version that we will study (v4) are 32 bits.
- Actually these 32 bits are splitted into two parts:
  - Network part or NetID
  - Host part or HostID
- Solution to this problem:
  - If the destination @IP has the same NetID as mine, I understand that this host is in my data-link network.
  - HostID therefore identifies the host within each network.



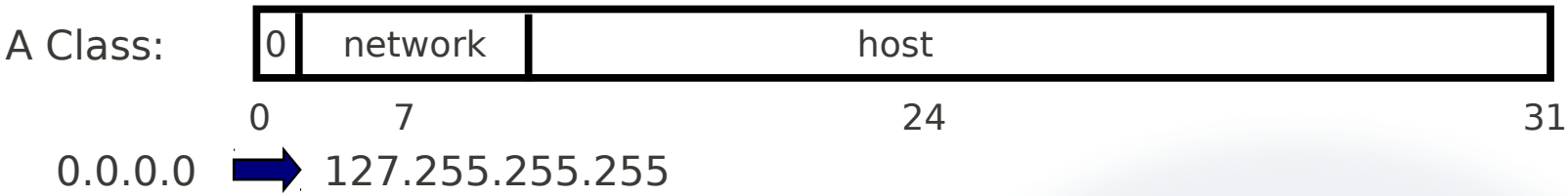
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Classes of Addresses

- There are five possible divisions or sorts of addresses (classes).
- Each sort defines which bits are used for NetID and which ones for the HostID.
- We will use the notation *w.x.y.z* to designate the values of the four bytes of a given IP address.
- The value of the first byte (*w*) points out the class that address belongs to.

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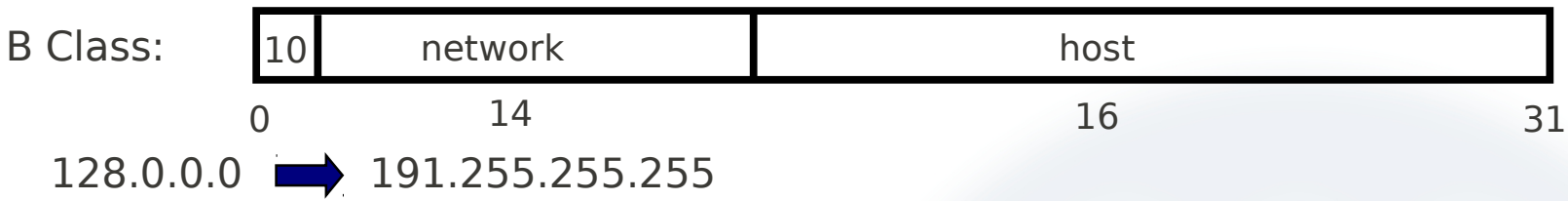
Class-A addresses



- *w* between 0 and 127.
- Network identifier (NetID): *w*.
- Host identifier (HostID): *x.y.z*.
- 126 Class-A networks with 16.777.214 hosts.
- Huge networks.
- Network *10* is private: 10.0.0.0 - 10.255.255.255
- Network *127* is internal: 127.0.0.0 - 127.255.255.255 (loopback)

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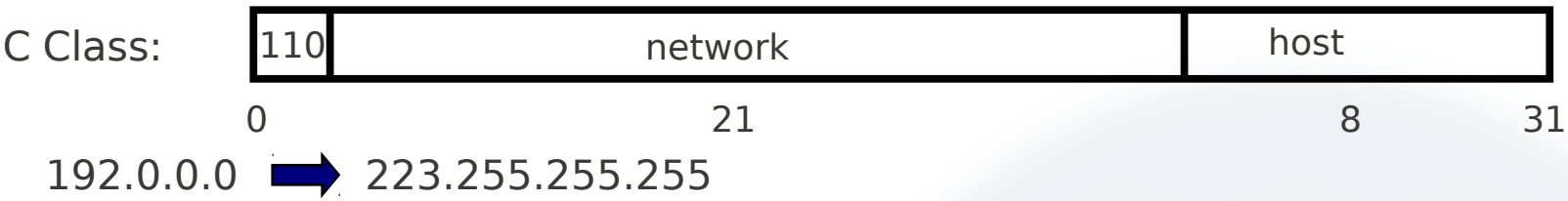
Class-B addresses



- *w* between 128 an 191
- NetID *w.x*
- HostID *y.z*
- 16.384 Class-B networks with 65.534 hosts.
- Moderate/Large networks.
- Private addresses: 172.16.0.0 - 172.31.255.255

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Class-C addresses

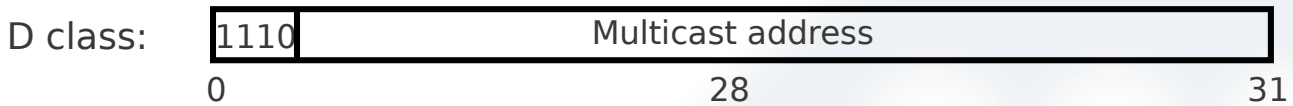


- *w* between 192 an 223
- Network identifier *w.x.y*
- Host identifier *z*
- 2.097.152 Class-C networks with 254 hosts.
- Small networks.
- Private addresses: 192.168.0.0 - 192.168.255.255.

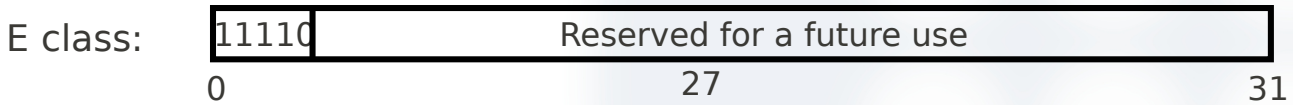
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D and E Classes

- Class-D & Class-E



224.0.0.0 ➡ 238.255.255.255
multicast addresses



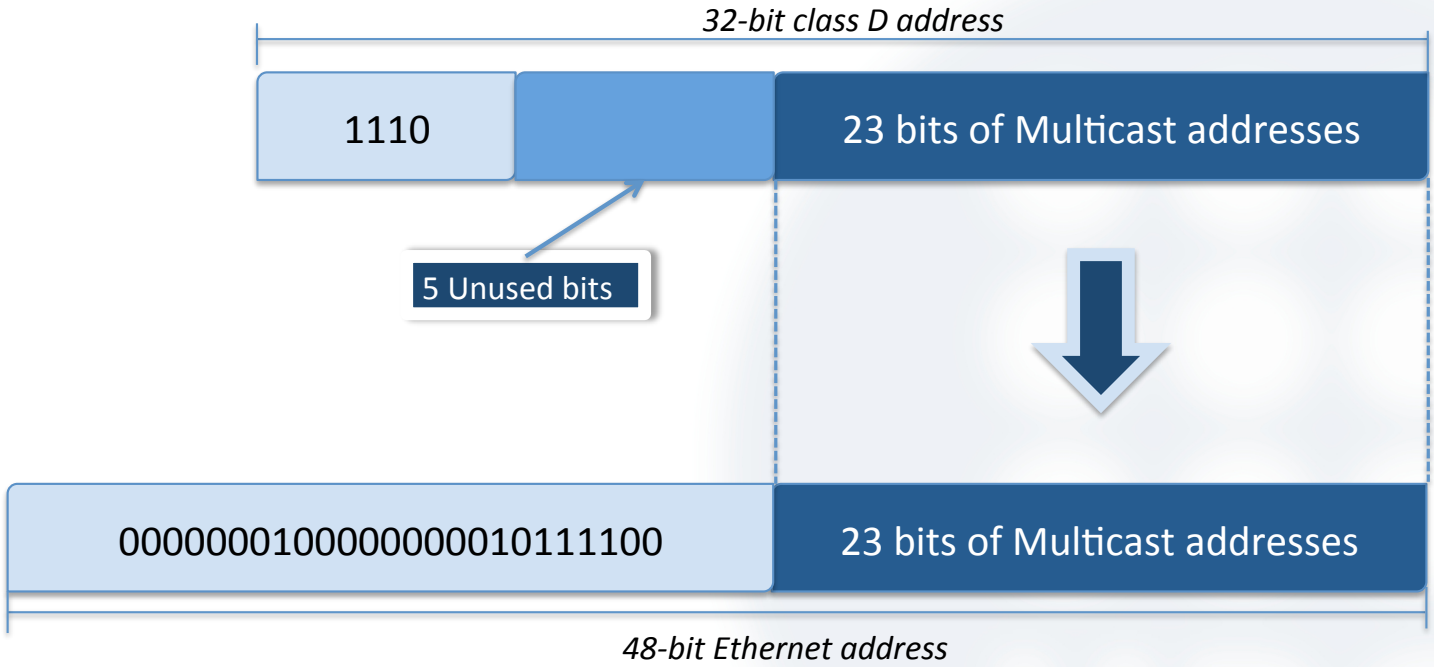
240.0.0.0 ➡ 247.255.255.255
reserved addresses

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Mapping class D to Ethernet

To encapsulate a m/c IP packet into layer 2 frame the physical address must be resolved. ARP is not feasible.

`m/c IP address is mapped into m/c MAC address by simple mapping`



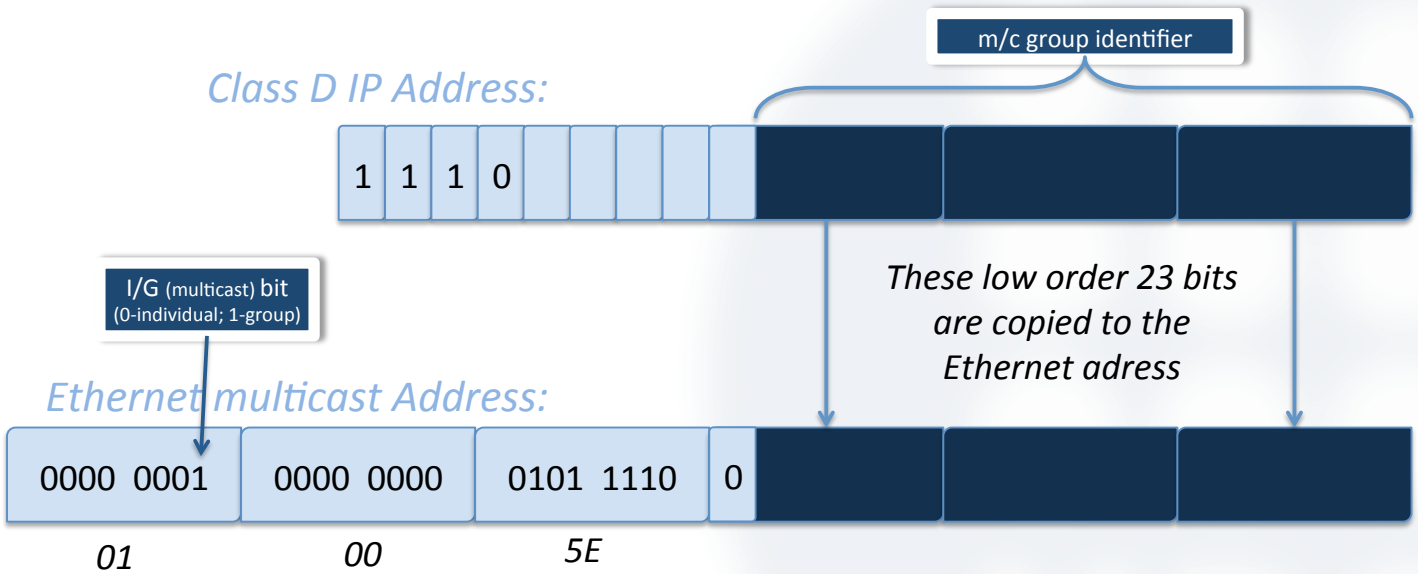
Most of LANs support m/c addressing. Ethernet included.

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Multicast Address Translation

In order to avoid usage of ARP in multicasting, the address translation is done by simple mapping:

The IP m/c group identifier is mapped onto the 23 least significant bits of the MAC address:

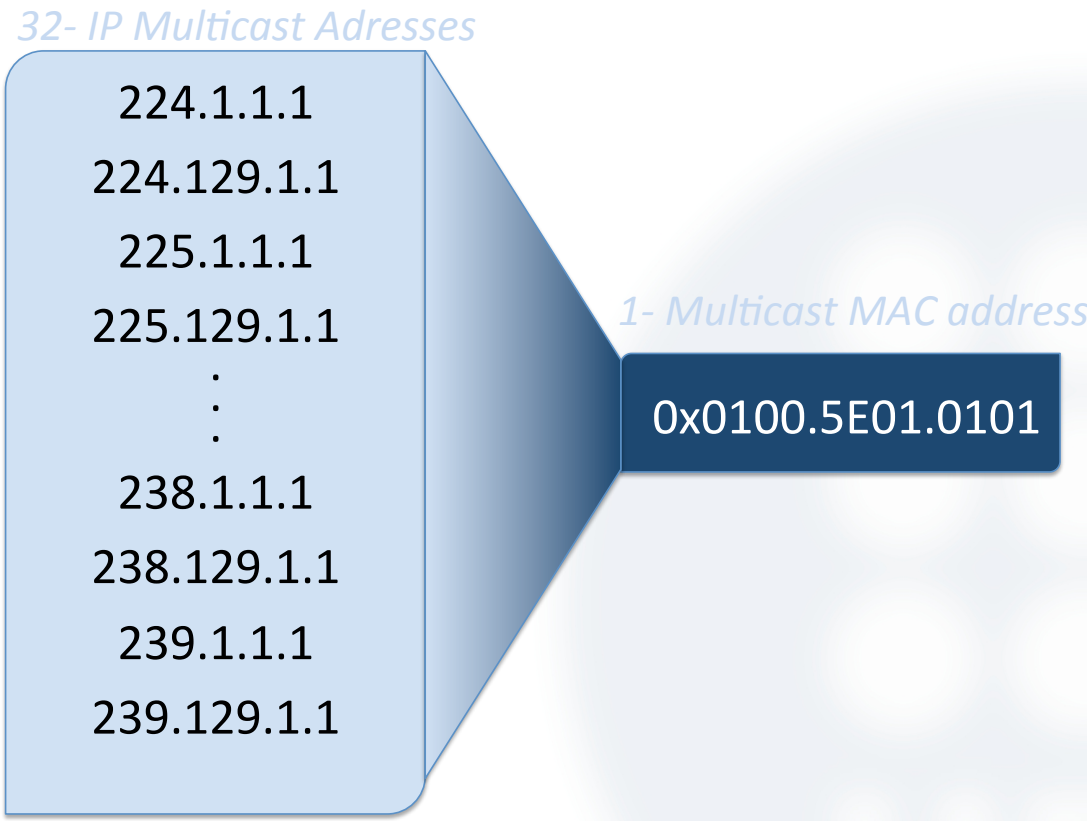


NOTE: There can be 32 IP address mapped to the same MAC address. Collisions are not very likely however.

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Ethernet addresses overlapping

The mapping is many-to-one because there are 28-23=5 bits lost

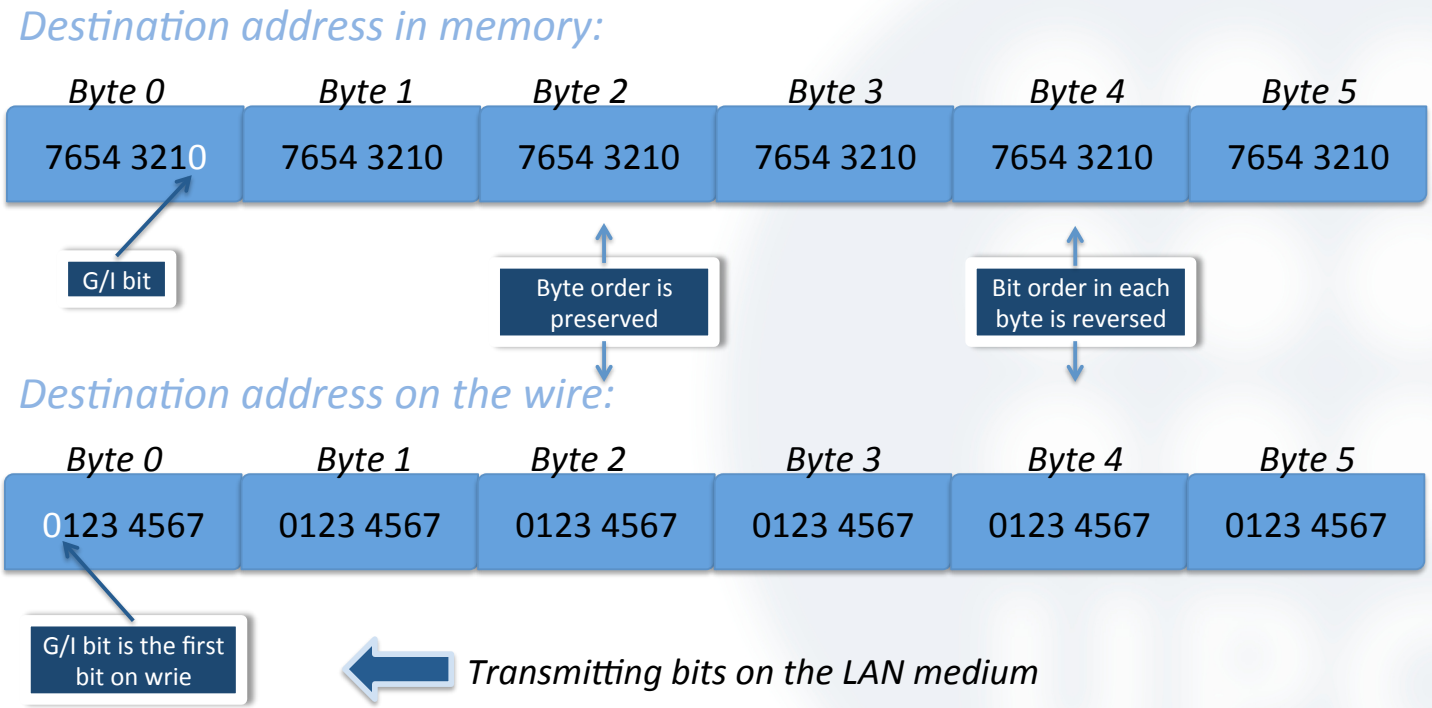


The host must check the m/c IP address and discard the packet if the IP addresses is not registered.

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Ethernet bit order

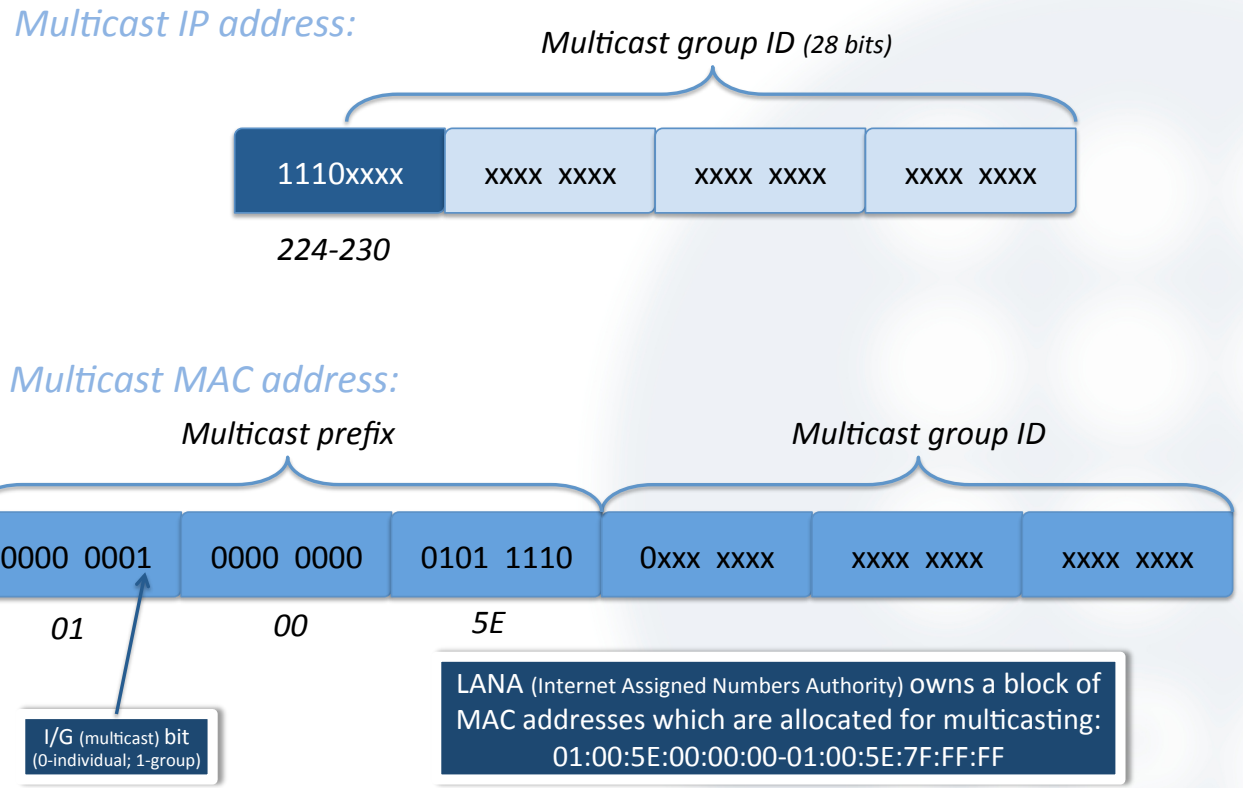
There is often a confusion about the position of the I/G bit in destination address, because the bit order of DA in memory and bit order during the frame transmission is different.



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Multicast Addresses I

Each address space, IP and physical address space, define multicast addresses.



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Multicast Addresses II

Assigned m/c Addresses (Category Addresses)

Address	Group
224.0.0.0	Reserved
224.0.0.1	All systems on this subnet
224.0.0.2	All routers on this subnet
224.0.0.4	DVMRP routers
224.0.0.5	OSPFIGP all routers
224.0.0.6	OSPFIGP designated routers
224.0.0.7	ST routers
224.0.0.8	ST hosts
224.0.0.9	RIP2 routers
224.0.0.10	IGRP routers
224.0.0.11	Mobile Agents

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Multicast Addresses III

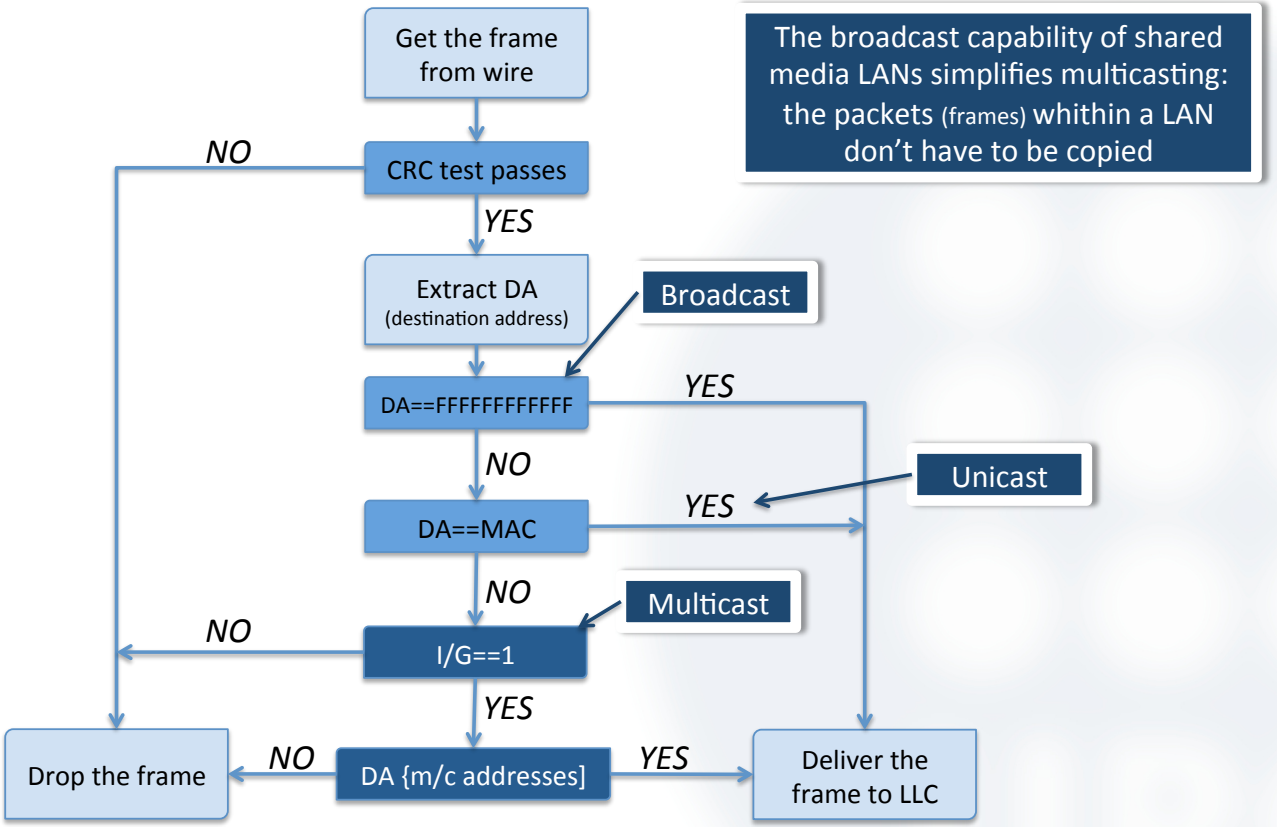
Assigned m/c Addresses (Conferencing)

Address	Group
224.0.1.7	AUDIONEWS
224.0.1.10	IETF-1-LOW-AUDO
224.0.1.11	IETF-1-AUDIO
224.0.1.12	IETF-1-VIDEO
224.0.1.13	IETF-2-LOW-AUDIO
224.0.1.14	IETF-2-AUDIO
224.0.1.15	IETT-2-VIDEO
224.0.1.16	MUSIC-SERVICE
224.0.1.17	SEANET-TELEMETRY
224.0.1.18	SEANET-IMAGE

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Multicast within Ethernet

How the MAC layer of the Ethernet interface works:



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Multicast Scoping

Multicasts are flooded across MAC-layer bridges along a spanning tree

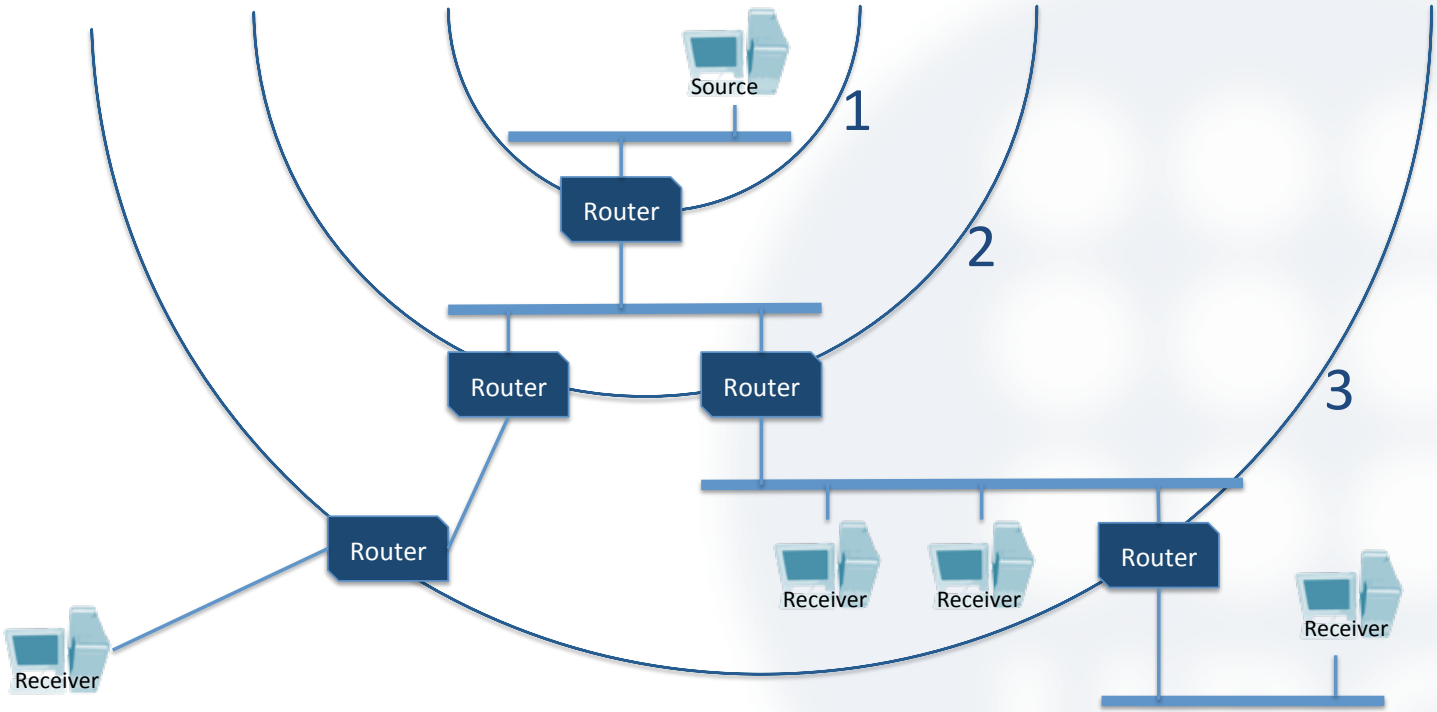
Scope: How far do transmissions propagate?

- Implicit scoping: Reserved Mcast addresses => do not leave subnet.
- TTL-based scoping:
  - Multicast routers have a configured TTL threshold
  - Multicast datagram dropped if TTL <= TTL threshold
- Administrative scoping:
  - Use a portion of class D address space (239.0.0.0 thru 239.255.255.255)
  - Truly local to admin domain; address reuse possible.

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Multicast Scope control

- Multicast Scope Control – Small TTLs



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Administrative Scope

- According to RFC 2365: 239.0.0.0–239.255.255.255
- Private address space
  - Similar to Address Allocation for Private Internets of unicast addresses (RFC 1918)
  - Not used for global Internet traffic
  - Used to limit scope of multicast traffic
  - Same addresses may be in use at different locations for different multicast sessions
- Examples:
  - Site-local scope: 239.253.0.0/16
  - Organization-local scope: 239.192.0.0/14

Review: RFC 5771 and
http://www.iana.org/assignments/multicast-addresses/multicast-addresses.xml

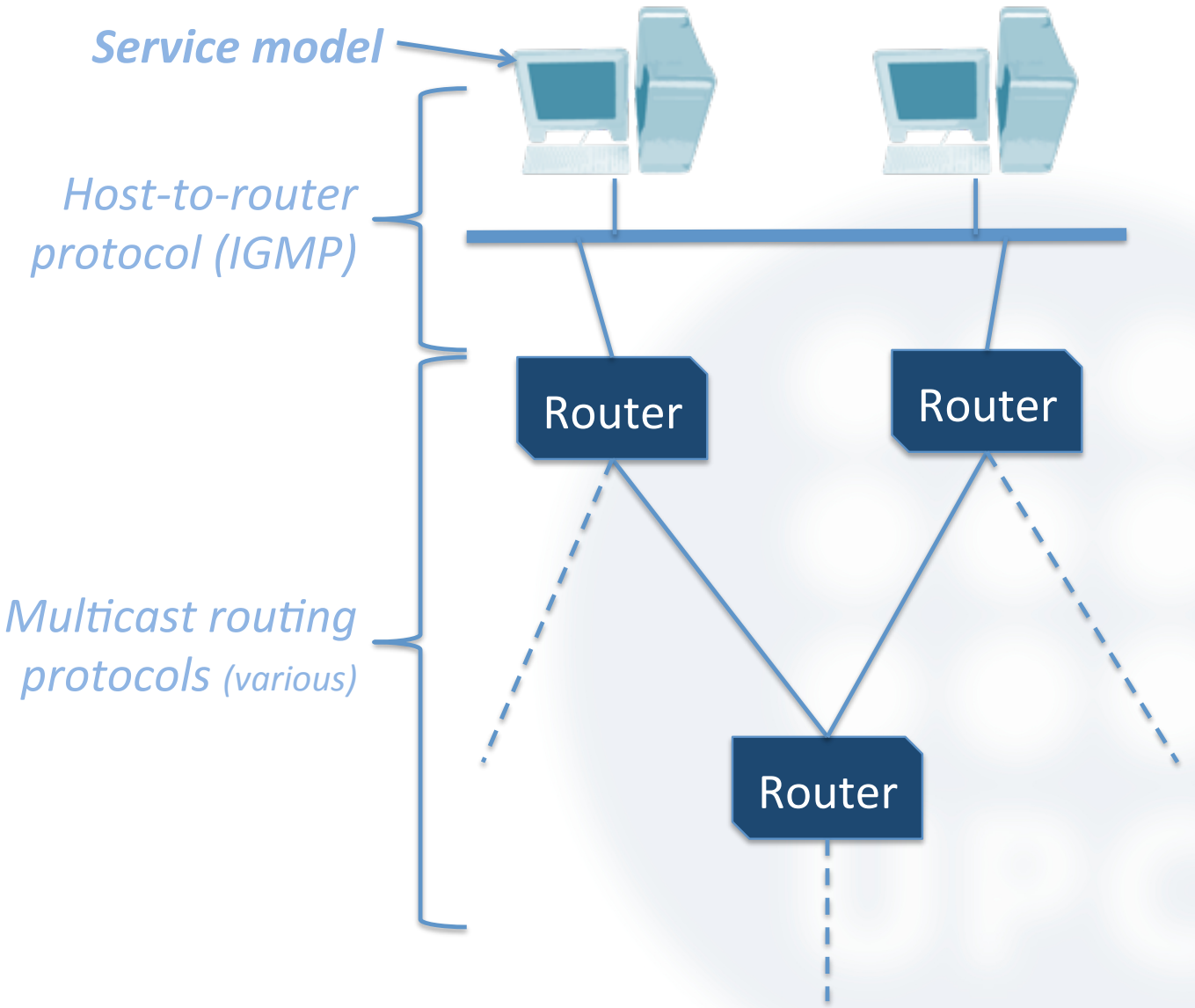
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IGMP

- IGMP is used by IP hosts to register their dynamic multicast group membership.
- It is also used by connected routers to discover these group members.

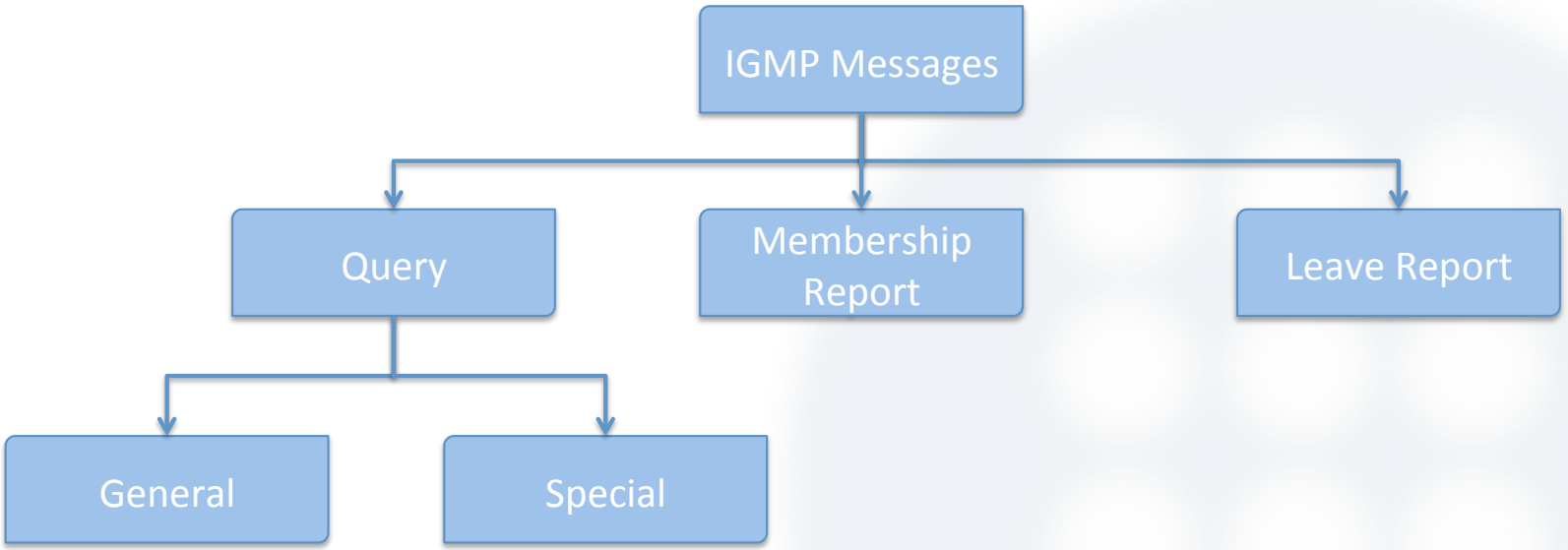


IGMP versions:

- IGMP v1: RFC 1112, 1989
- IGMP v2: RFC 2236, 1997
- IGMP v3: RFC 3376, 2002

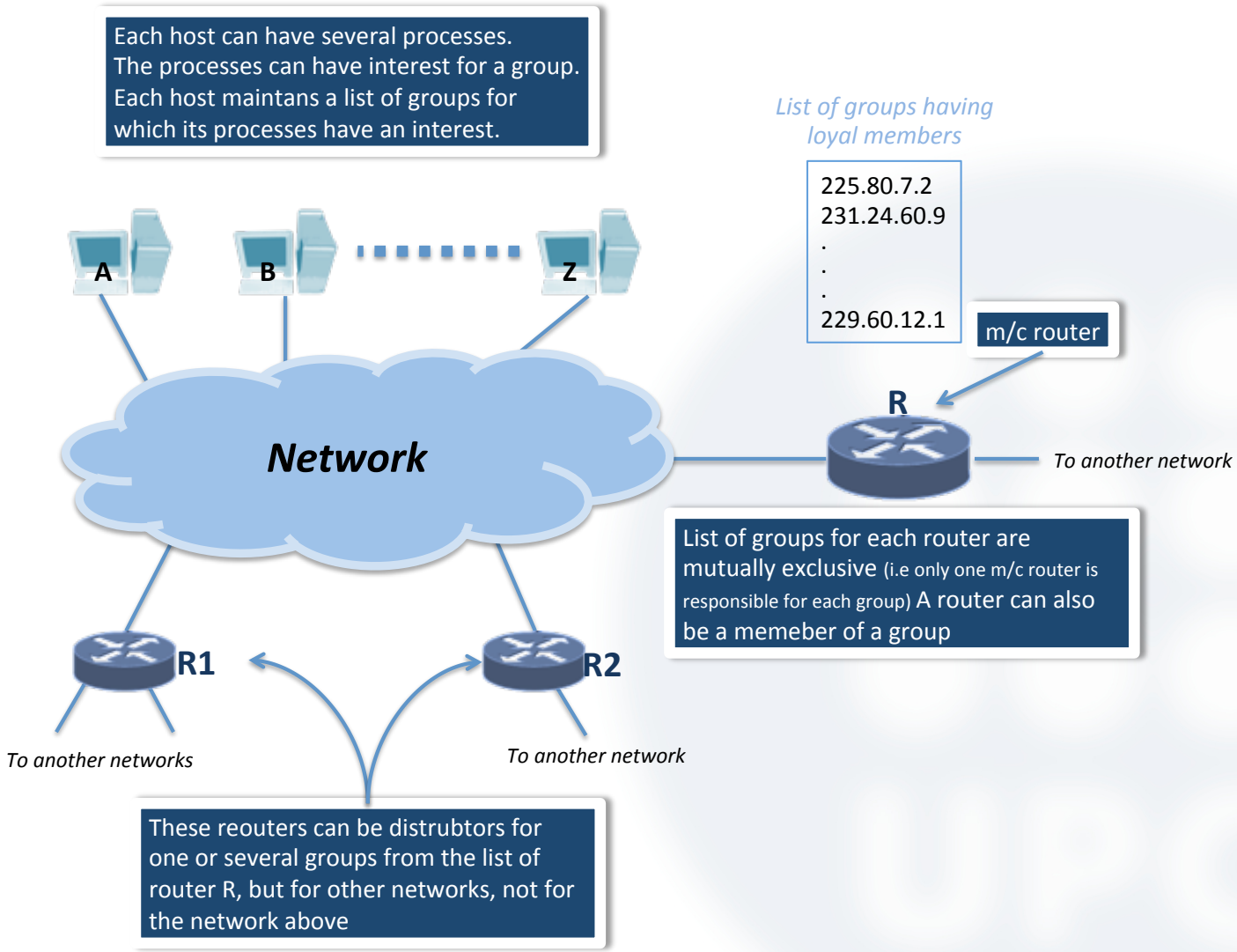
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IGMP Message Types



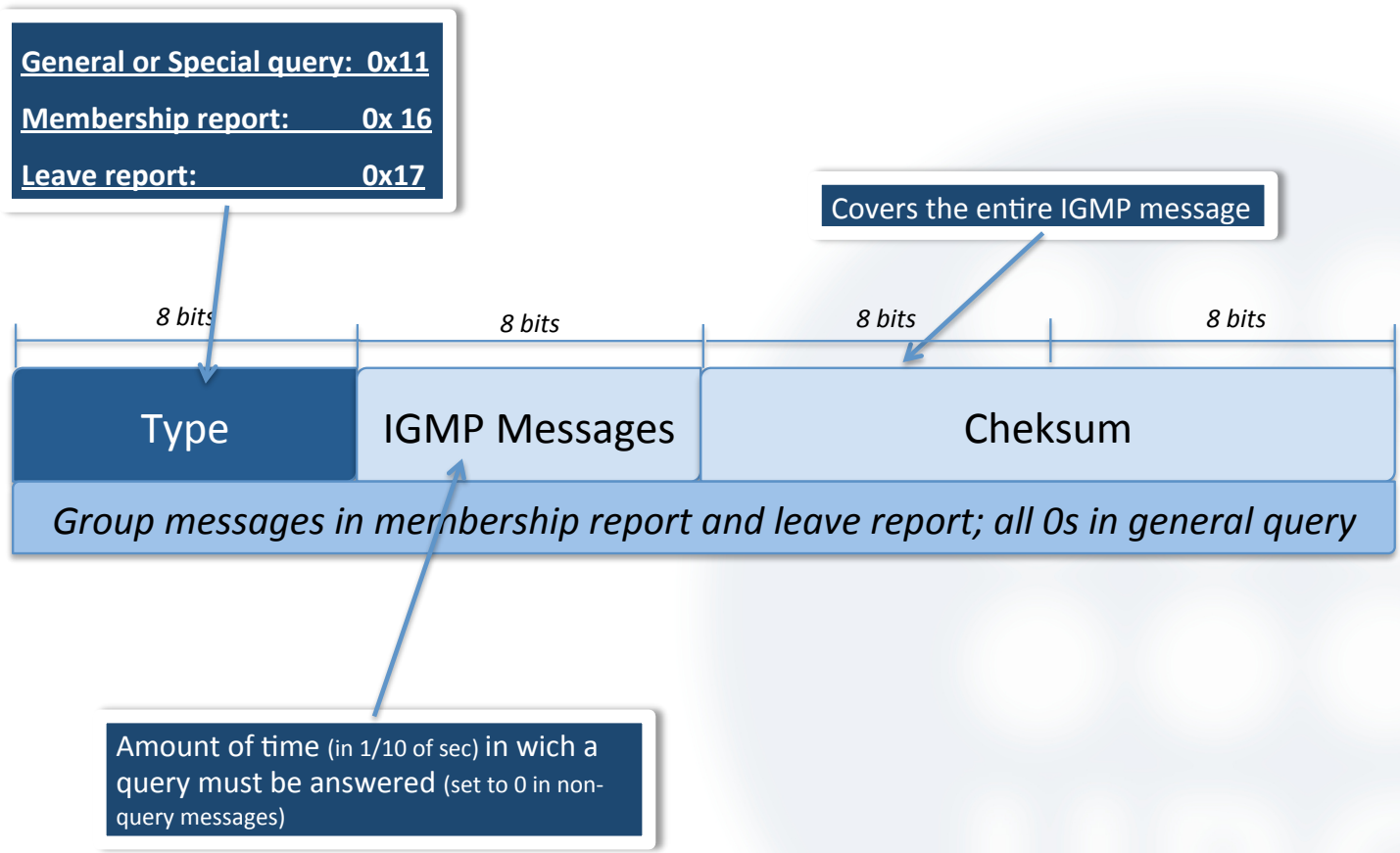
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IGMP Message Format



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IGMP Operation



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Membership Report

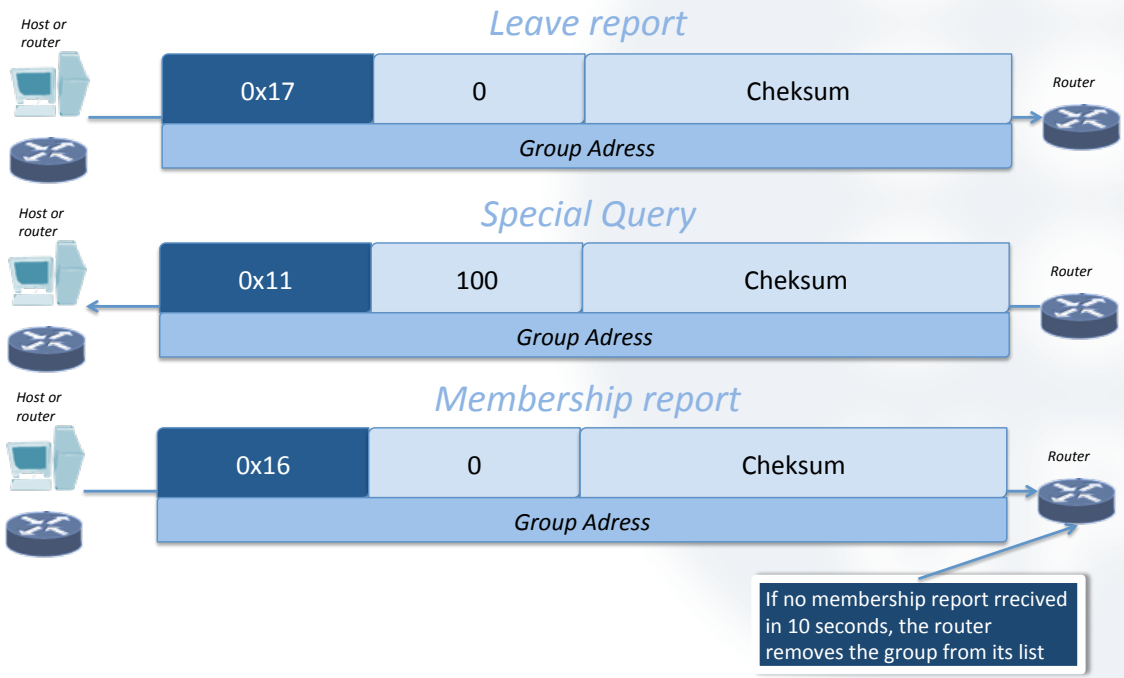
- A host or a router can join a group.
- A host maintains a list of processes that have interest in a group.
- Join operation:
  - When a process wants to join a new group, it sends its request to the host.
  - The host then adds the name of the process and the name of the requested group to its list
  - The host sends the membership report to the router. The report can be sent twice (fiability purpose)



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Leave Report

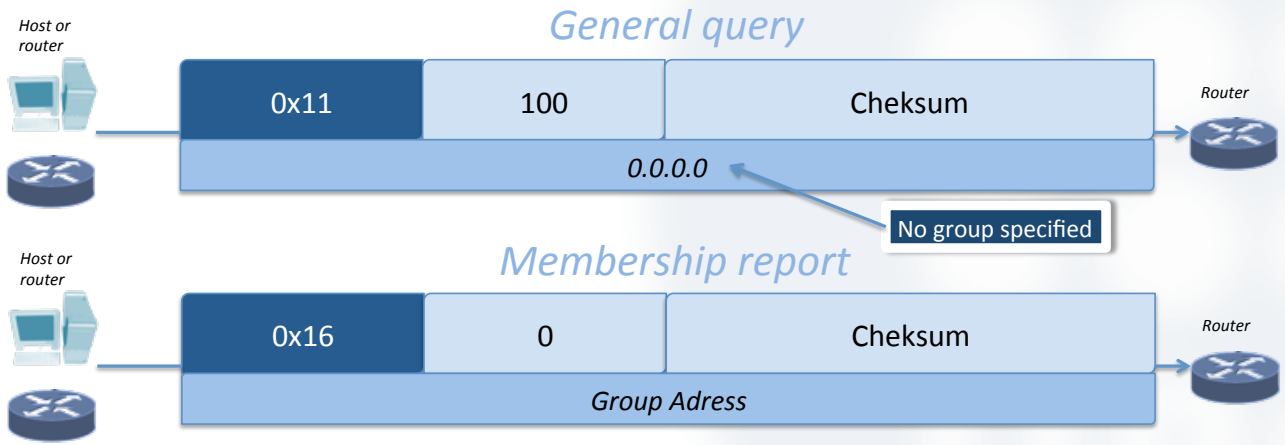
- When a host sees that no process is interested in a specific group G, it sends a leave report.
- If a router receives a leave report it will not purge the list if there are still other hosts interested in that group.
- For that purpose the router sends a special query message with a specified response time for the group in question to see if there is anyone interested in that group.
- If there is no response a membership report, it purges the list.



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General Query Message

Membership report and leave report are not enough to maintain the membership information.
Example: a host that is a member of a group can shut down and the m/c router would never receive the leave report.
Therefore the m/c router monitors the hosts and routers in the LAN by periodically sending (by default every 125 sec) general query message.
Hosts/routers respond by sending membership reports if they still have interest in those groups.



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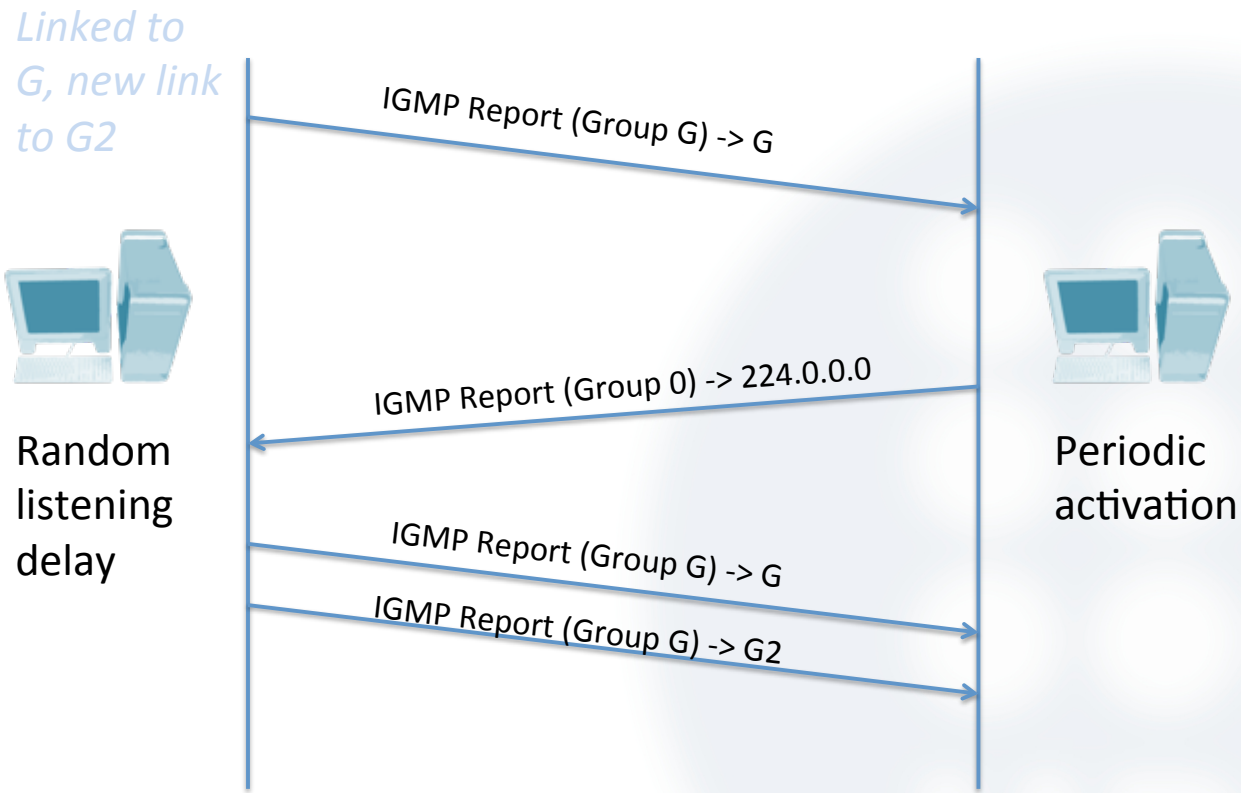
Delayed Response

How can be made sure that only one host answers the query, while the others which have to report the same group, or groups are not?
This is achieved with delayed response:

- To keep the traffic low the response to a general query message must be done by only one host for a given group.
- When a host receives a general query message it delays the response: it sets a timer for each group to a different random value between 0 and 10 seconds
- If the host receives a response from another host, whose timer for that group has expired earlier, the host cancels its response
- Only one router on the LAN is designated for sending the query messages (the querier router)

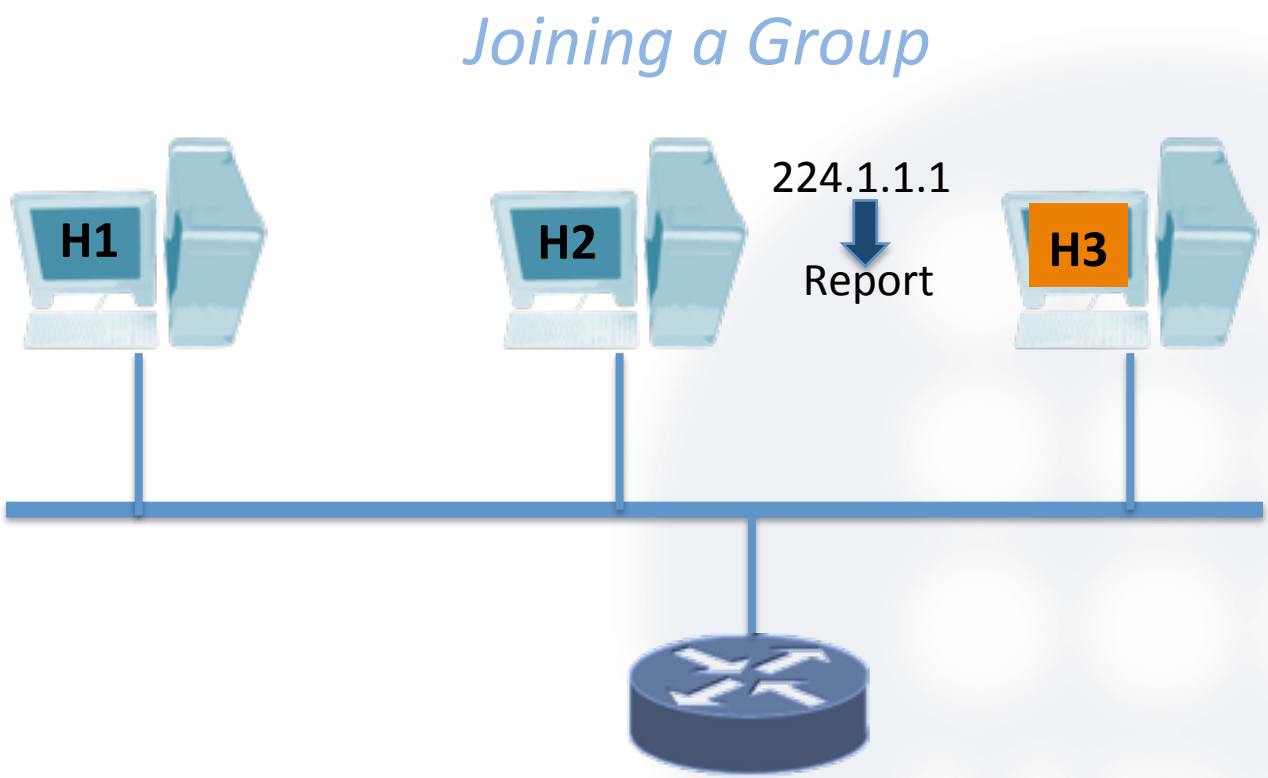
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IGMPv1



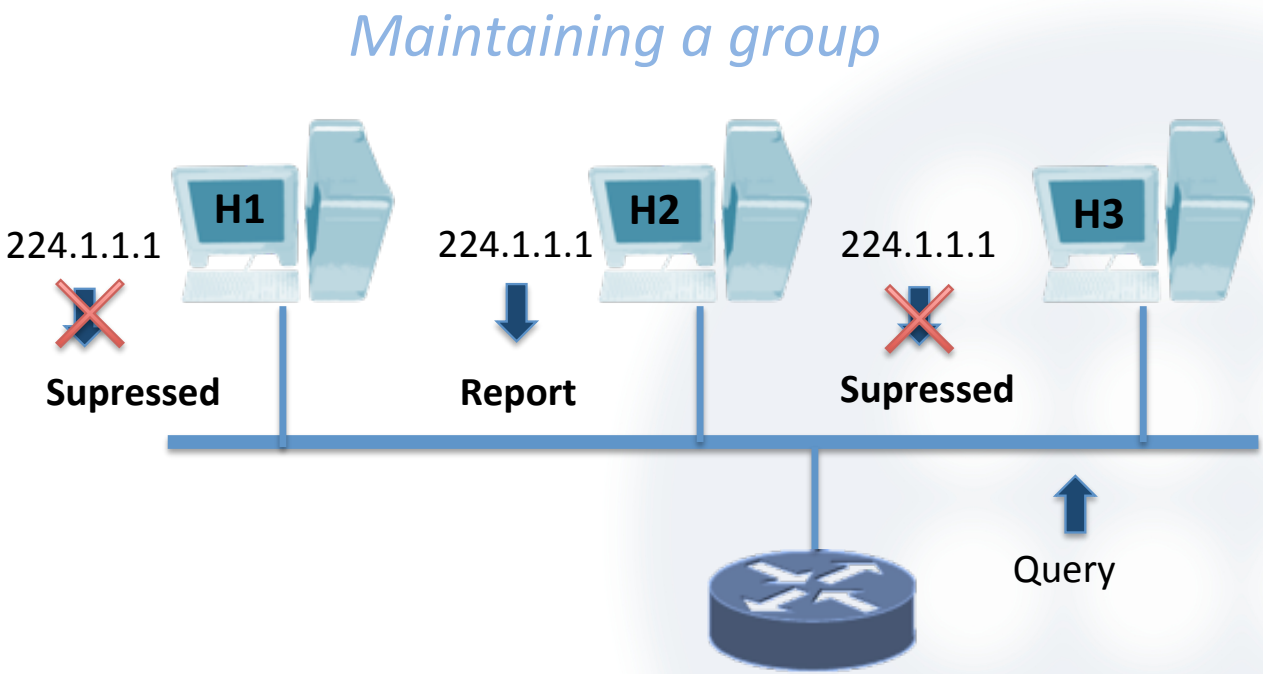
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IGMP Joinnning



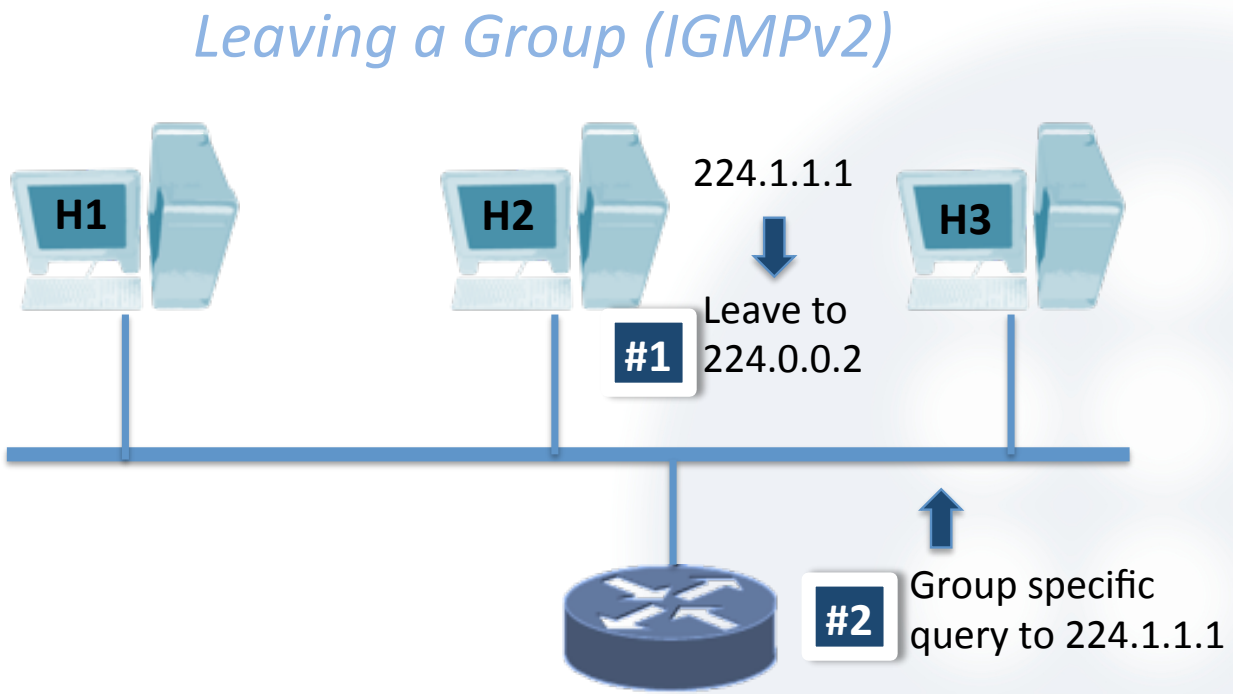
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IGMP Maintaining



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IGMPv2 Leaving



Example 1

A query message was received at time 0; the random delay time (in tenths of seconds) for each group is shown next to the group address. Show the sequence of report messages.

Group	Timer
225.14.0.0	30
228.42.0.0	12
230.43.0.0	80

Group	Timer
225.42.0.0	48
251.71.0.0	50

Group	Timer
225.14.0.0	62
228.43.0.0	70



Solution:

Group	Timer	From Host
12	228.42.0.0	A
30	225.14.0.0	A
50	251.71.0.0	B
70	230.43.0.0	C

Four messages are sent instead of seven messages.

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IGMPv3

- Adds Include/Exclude Source Lists
- Host listen to a subset of host sending to a group
- Requieres new API for applications
- New IGMP stack in the OS
- Applications need to include new features Include/Exclude

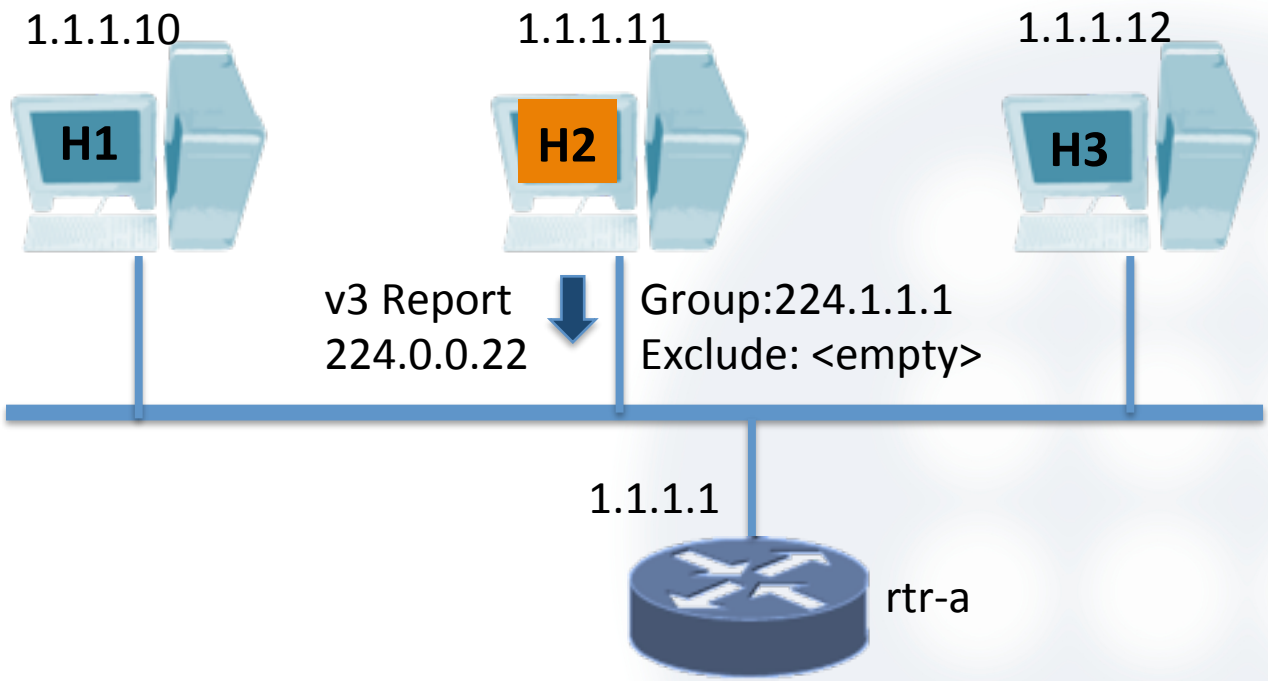
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IGMPv3

- New Membership Report Address: 224.0.0.22 (only IGMPv3 routers listen)
- IGMPv3 hosts send reports to this address instead of the tarjet group.
- No report suppression: All host on wire respond to Queries
- Useful for large of host in the subnet

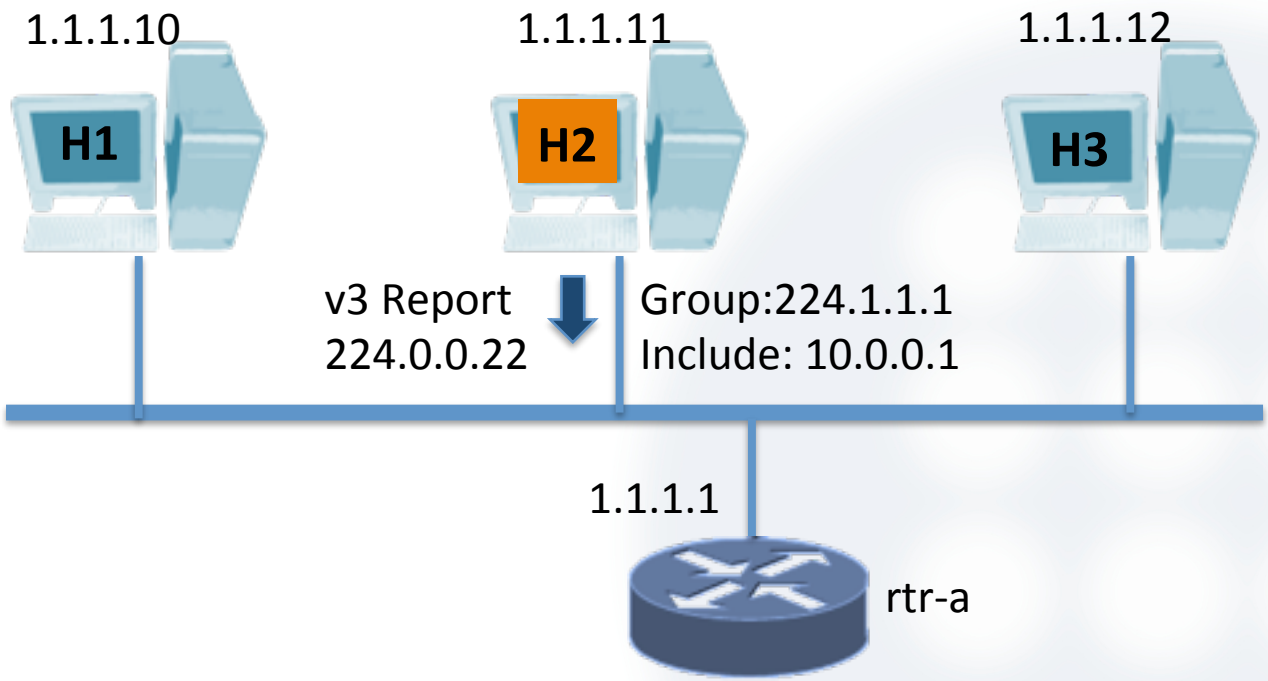
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IGMPV3 Joinnning



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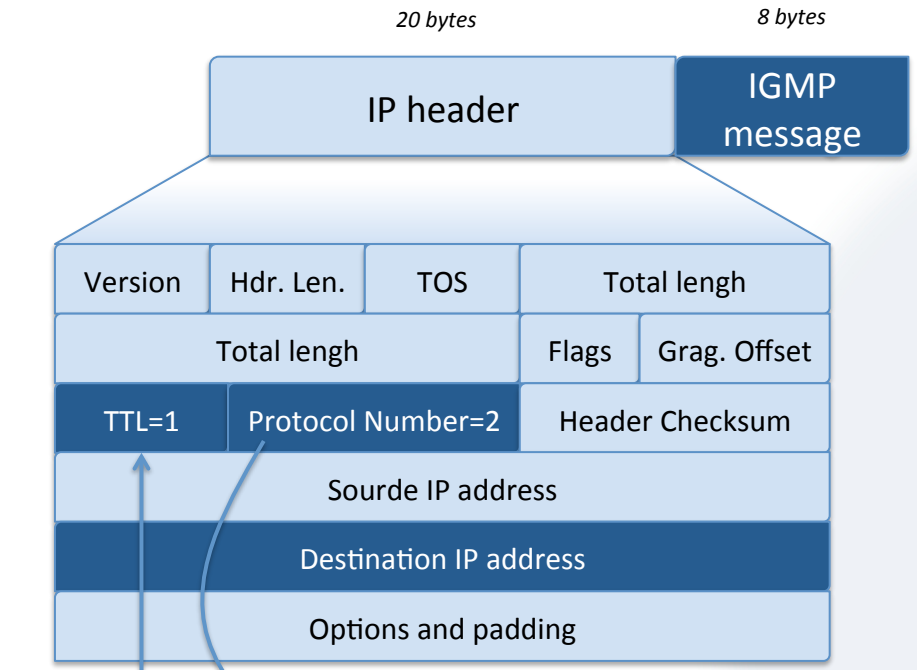
IGMPV3 Joinnning specific



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Encapsulation of IGMP Messages

IGMP messages are encapsulated in IP datagrams



All systems on this subnet

Query	224.0.0.1.
Membership Report	m/c addr. of the group
Leave report	224.0.0.2

All routers on this subnet

IGMP message doesn't leave the LAN

Prot. Num	Protocol	Description
1	ICMP	Internet Control Message
2	IGMP	Internet Group Management
3	GGP	Gateway-to-Gateway
4	IP	IP in IP (encapsulation)

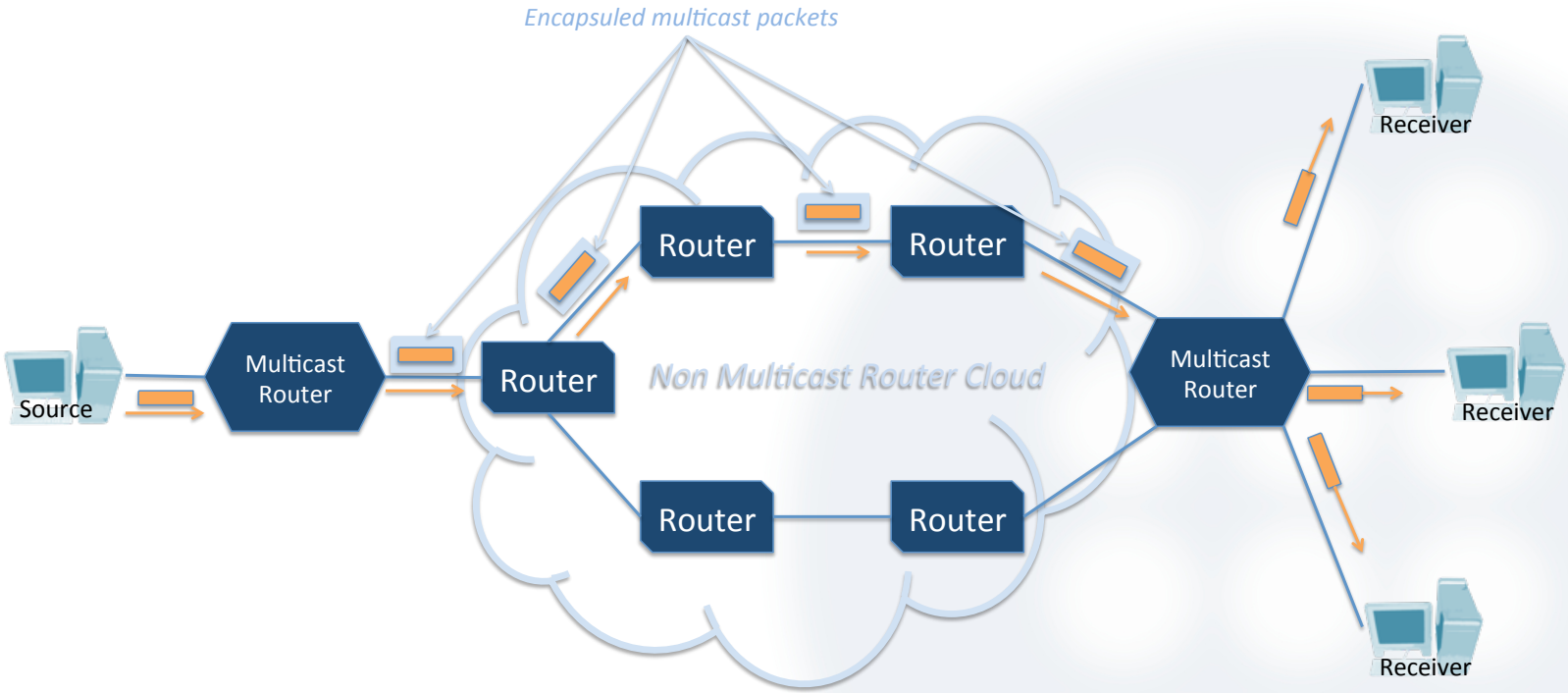
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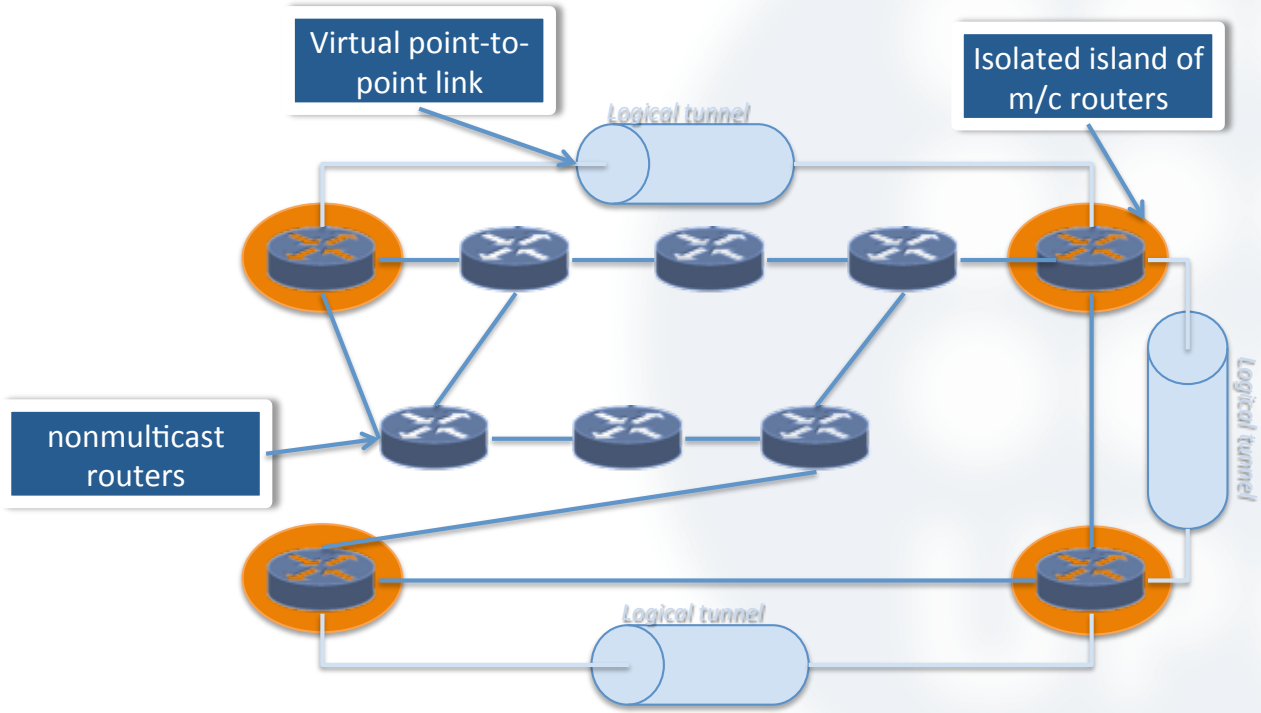
MBONE architecture



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Multicast Backbone (MBONE)

- Multimedia and real-time applications have increased the need for multicasting in Internet.
- Only a small fraction of Internet routers support m/c.
- The solution is to use tunneling: provides a logical connection of m/c routers through non-multicast routers, called multicast backbone.



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Multicast vs Unicast Routing

Multicast routing is different from unicast routing:

- Unicast routing:
  - Unicast routing is concerned about where the packet is going
  - Unicast forwarding: destination IP address directly indicates where to forward the packet
  - Forwarding is hop-by-hop
  - The unicast routing table determines the interface and the nexthop router to forward the packet
- Multicast routing:
  - Multicast routing is concerned about where the packet came from
  - The destination IP address (group) does not directly indicate where to forward the packet
  - Receivers must be “connected” to the tree before the traffic begins to flow

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Multicast Routing I

Multicast trees are spanning trees

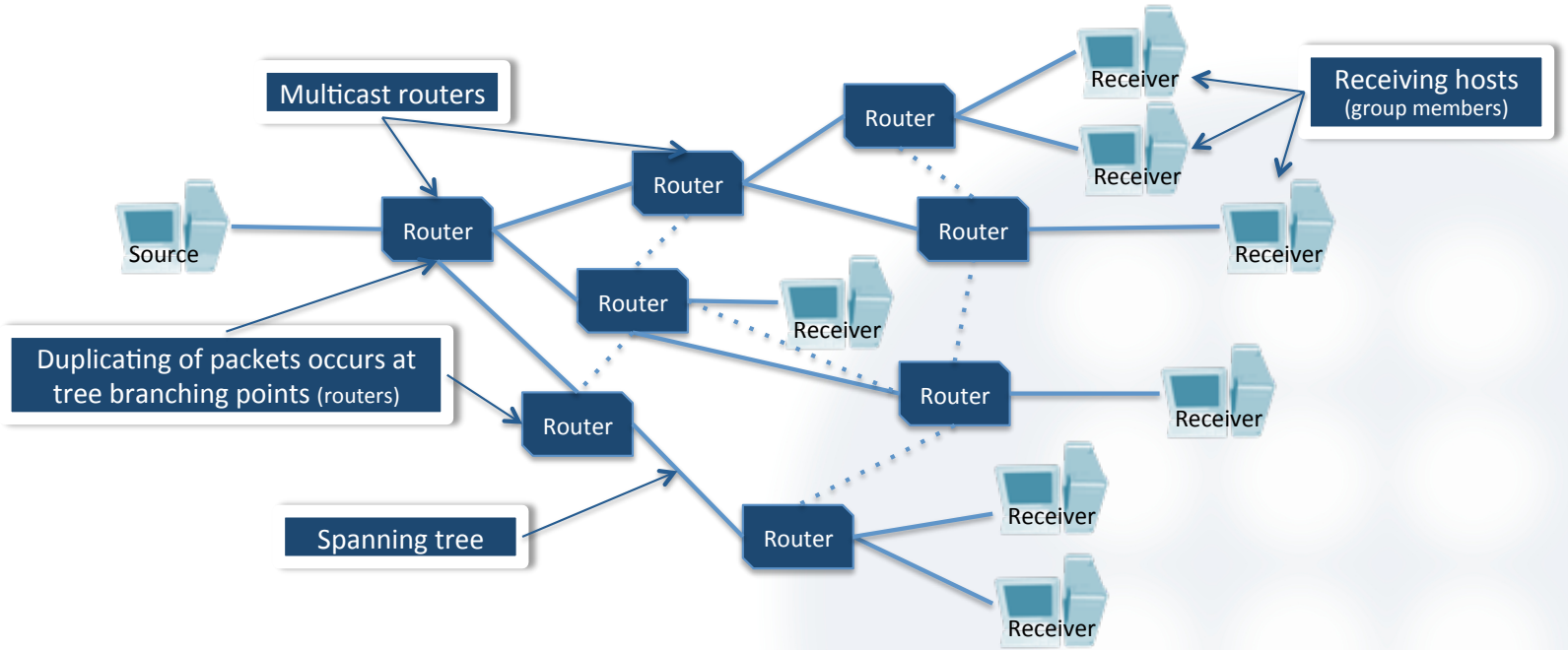
The source is located at the root of the tree, and the group members are the leaves of the tree.

Basic requirements:

- Every group member should receive only one copy of the m/c packet
- Non members must not receive a copy
- A m/c packet must not be received by a router more than once (no loops)
- Paths from the source to each destination must be optimal (shortest path)

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Multicast Routing II



The optimal tree is called shortest path spanning tree.

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Types of multicast trees

- Source-based trees: each source has a tree that spans a multicast group
- Shared trees: single distributed tree shared among all sources (Common Node RP, Rendezvous Point)